

```
import pandas as pd
from google.colab import files
uploaded = files.upload()
```

No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

Saving zomato.csv.zip to zomato.csv.zip

```
!unzip zomato.csv.zip
```

```
Archive: zomato.csv.zip
  inflating: zomato.csv
```

```
!!ls
```

```
sample_data  zomato.csv  zomato.csv.zip
```

```
df = pd.read_csv("zomato.csv")
```

```
df.head()
```

	address	name	online_order	book_table	rate	votes	phone	location	rest_type	dish_liked	cuisines
0	942, 21st Main Road, 2nd Stage, Banashankari, ...	Jalsa	Yes	Yes	4.1/5	775	080 42297555\r\n+91 9743772233	Banashankari	Casual Dining	Pasta, Lunch Buffet, Masala Papad, Paneer Laja...	Mughlai, Chinese
1	2nd Floor, 80 Feet Road, Near Big Bazaar, 6th ...	Spice Elephant	Yes	No	4.1/5	787	080 41714161	Banashankari	Casual Dining	Momos, Lunch Buffet, Chocolate Nirvana, Thai G...	Chinese, Indian
2	1112, Next to KIMS Medical College, 17th Cross...	San Churro Cafe	Yes	No	3.8/5	918	+91 9663487993	Banashankari	Cafe, Casual Dining	Churros, Cannelloni, Minestrone Soup, Hot Choc...	Mexican, Italian
3	1st Floor, Annakuteera, 3rd Stage, Banashankar...	Addhuri Udupi Bhojana	No	No	3.7/5	88	+91 9620009302	Banashankari	Quick Bites	Masala Dosa	South Indian, North Indian

```
df.shape
```

```
(56252, 13)
```

```
df.columns
```

```
Index(['address', 'name', 'online_order', 'book_table', 'rate', 'votes',
       'phone', 'location', 'rest_type', 'dish_liked', 'cuisines',
       'approx_cost(for two people)', 'listed_in(type)'],
      dtype='object')
```

```
df.isnull().sum()
```

	0
address	17
name	16
online_order	19
book_table	58
rate	7838
votes	78
phone	1296
location	126
rest_type	338
dish_liked	28225
cuisines	203
approx_cost(for two people)	521
listed_in(type)	4610

dtype: int64

df.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 56252 entries, 0 to 56251
Data columns (total 13 columns):
#   Column              Non-Null Count  Dtype
---  -
0   address             56235 non-null  object
1   name                56236 non-null  object
2   online_order        56233 non-null  object
3   book_table          56194 non-null  object
4   rate                48414 non-null  object
5   votes               56174 non-null  object
6   phone               54956 non-null  object
7   location            56126 non-null  object
8   rest_type           55914 non-null  object
9   dish_liked          28027 non-null  object
10  cuisines             56049 non-null  object
11  approx_cost(for two people) 55731 non-null  object
12  listed_in(type)      51642 non-null  object
dtypes: object(13)
memory usage: 5.6+ MB
```

df.head()

	address	name	online_order	book_table	rate	votes	phone	location	rest_type	dish_liked	cuisines
0	942, 21st Main Road, 2nd Stage, Banashankari, ...	Jalsa	Yes	Yes	4.1/5	775	080 42297555\n+91 9743772233	Banashankari	Casual Dining	Pasta, Lunch Buffet, Masala Papad, Paneer Laja...	Mughlai, Chinese
1	2nd Floor, 80 Feet Road, Near Big Bazaar, 6th ...	Spice Elephant	Yes	No	4.1/5	787	080 41714161	Banashankari	Casual Dining	Momos, Lunch Buffet, Chocolate Nirvana, Thai G...	Chinese, Indian
2	1112, Next to KIMS Medical College, 17th Cross...	San Churro Cafe	Yes	No	3.8/5	918	+91 9663487993	Banashankari	Cafe, Casual Dining	Churros, Cannelloni, Minestrone Soup, Hot Choc...	Mexican, Italian
3	1st Floor, Annakuteera, 3rd Stage, Banashankar...	Addhuri Udupi Bhojana	No	No	3.7/5	88	+91 9620009302	Banashankari	Quick Bites	Masala Dosa	South Indian

```
df.duplicated().sum()
```

```
np.int64(15703)
```

```
df.drop_duplicates(inplace=True)
```

```
df.shape
```

```
(40549, 13)
```

```
df.isnull().sum()
```

	0
address	2
name	1
online_order	4
book_table	38
rate	4145
votes	58
phone	808
location	98
rest_type	251
dish_liked	18226
cuisines	151
approx_cost(for two people)	344
listed_in(type)	2907

```
dtype: int64
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 40549 entries, 0 to 56251
Data columns (total 13 columns):
#   Column              Non-Null Count  Dtype
---  -
0   address             40547 non-null  object
1   name                40548 non-null  object
2   online_order        40545 non-null  object
3   book_table          40511 non-null  object
4   rate                36404 non-null  object
5   votes               40491 non-null  object
6   phone               39741 non-null  object
7   location            40451 non-null  object
8   rest_type           40298 non-null  object
9   dish_liked          22323 non-null  object
10  cuisines             40398 non-null  object
11  approx_cost(for two people) 40205 non-null  object
12  listed_in(type)      37642 non-null  object
dtypes: object(13)
memory usage: 4.3+ MB
```

```
df.head()
```

	address	name	online_order	book_table	rate	votes	phone	location	rest_type	dish_liked	cuisi
0	942, 21st Main Road, 2nd Stage, Banashankari, ...	Jalsa	Yes	Yes	4.1/5	775	080 42297555\n+91 9743772233	Banashankari	Casual Dining	Pasta, Lunch Buffet, Masala Papad, Paneer Laja...	Mug Chi
1	2nd Floor, 80 Feet Road, Near Big Bazaar, 6th ...	Spice Elephant	Yes	No	4.1/5	787	080 41714161	Banashankari	Casual Dining	Momos, Lunch Buffet, Chocolate Nirvana, Thai G...	Chir M In
2	1112, Next to KIMS Medical College, 17th Cross...	San Churro Cafe	Yes	No	3.8/5	918	+91 9663487993	Banashankari	Cafe, Casual Dining	Churros, Cannelloni, Minestrone Soup, Hot Choc...	C Mex It
3	1st Floor, Annakuteera, 3rd Stage, Banashankar...	Addhuri Udupi Bhojana	No	No	3.7/5	88	+91 9620009302	Banashankari	Quick Bites	Masala Dosa	S In M Ir

```
df['rate'].fillna(df['rate'].mode()[0], inplace=True)

df['location'].fillna('Unknown', inplace=True)

df['cuisines'].fillna('Unknown', inplace=True)

df['dish_liked'].fillna('Not Mentioned', inplace=True)

df['listed_in(type)'].fillna('Others', inplace=True)

df['approx_cost(for two people)'].fillna(
    df['approx_cost(for two people)'].mode()[0],
    inplace=True
)
```

/tmp/ipykernel_4950/3629716864.py:1: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are
For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col]

```
df['rate'].fillna(df['rate'].mode()[0], inplace=True)
/tmp/ipykernel_4950/3629716864.py:3: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through   
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are  
For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col]
```

```
df['location'].fillna('Unknown', inplace=True)
/tmp/ipykernel_4950/3629716864.py:5: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through   
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are  
For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col]
```

```
df['cuisines'].fillna('Unknown', inplace=True)
/tmp/ipykernel_4950/3629716864.py:7: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through   
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are  
For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col]
```

```
df['dish_liked'].fillna('Not Mentioned', inplace=True)
/tmp/ipykernel_4950/3629716864.py:9: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through   
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are  
For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col]
```

```
df['listed_in(type)'].fillna('Others', inplace=True)
/tmp/ipykernel_4950/3629716864.py:11: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through   
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are
```

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col]

```
df['approx_cost(for two people)'].fillna(
```

```
df.isnull().sum()
```

	0
address	2
name	1
online_order	4
book_table	38
rate	0
votes	58
phone	808
location	0
rest_type	251
dish_liked	0
cuisines	0
approx_cost(for two people)	0
listed_in(type)	0

dtype: int64

```
df['rate'].unique()
```

```
array(['4.1/5', '3.8/5', '3.7/5', ...,
      " but we weren't blown away. It was",
      " also a slightly larger portion size wud be much appreciated...\n\nService - 5/10\n\nReally needs a lot of\n tweak up... the waiters are never to be found on the floor... they lack knowledge of dishes... very uninviting\n overall...\n\nOverall not a grt experience :(')",
      ' small place '], dtype=object)
```

```
df.head(3)
```

	address	name	online_order	book_table	rate	votes	phone	location	rest_type	dish_liked	cuisines
0	942, 21st Main Road, 2nd Stage, Banashankari, ...	Jalsa	Yes	Yes	4.1/5	775	42297555/r/n+91 9743772233	Banashankari	Casual Dining	Pasta, Lunch Buffet, Masala Papad, Paneer Laja...	North Indian, Mughlai, Chinese
1	2nd Floor, 80 Feet Road, Near Big Bazaar, 6th ...	Spice Elephant	Yes	No	4.1/5	787	080 41714161	Banashankari	Casual Dining	Momos, Lunch Buffet, Chocolate Nirvana, Thai G...	Chinese, North Indian, Tha
	1112. Next to	-								Churros,	-

```
df.iloc[0]
```

```

0
address          942, 21st Main Road, 2nd Stage, Banashankari, ...
name              Jalsa
online_order      Yes
book_table        Yes
rate              4.1/5
votes             775
phone             080 42297555\n\n+91 9743772233
location          Banashankari
rest_type         Casual Dining
dish_liked        Pasta, Lunch Buffet, Masala Papad, Paneer Laja...
cuisines          North Indian, Mughlai, Chinese
approx_cost(for two people)  800
listed_in(type)   Buffet
dtype: object

```

```
print(df.columns)
```

```

Index(['address', 'name', 'online_order', 'book_table', 'rate', 'votes',
      'phone', 'location', 'rest_type', 'dish_liked', 'cuisines',
      'approx_cost(for two people)', 'listed_in(type)'],
      dtype='object')

```

```
df[['name', 'rate']].head(10)
```

	name	rate
0	Jalsa	4.1/5
1	Spice Elephant	4.1/5
2	San Churro Cafe	3.8/5
3	Addhuri Udupi Bhojana	3.7/5
4	Grand Village	3.8/5
5	Timepass Dinner	3.8/5
6	Rosewood International Hotel - Bar & Restaurant	3.6/5
7	Onesta	4.6/5
8	Penthouse Cafe	4.0/5
9	service was sluggish as they took too long to... \nTop floor	

```

top_locations = df['location'].value_counts().head(10)
print(top_locations)

```

```

location
BTM          2900
Whitefield   1941
HSR          1843
Indiranagar  1765
Marathahalli 1668
Koramangala 5th Block 1659
JP Nagar     1487
Jayanagar    1327
Electronic City 1211
Bellandur    1133
Name: count, dtype: int64

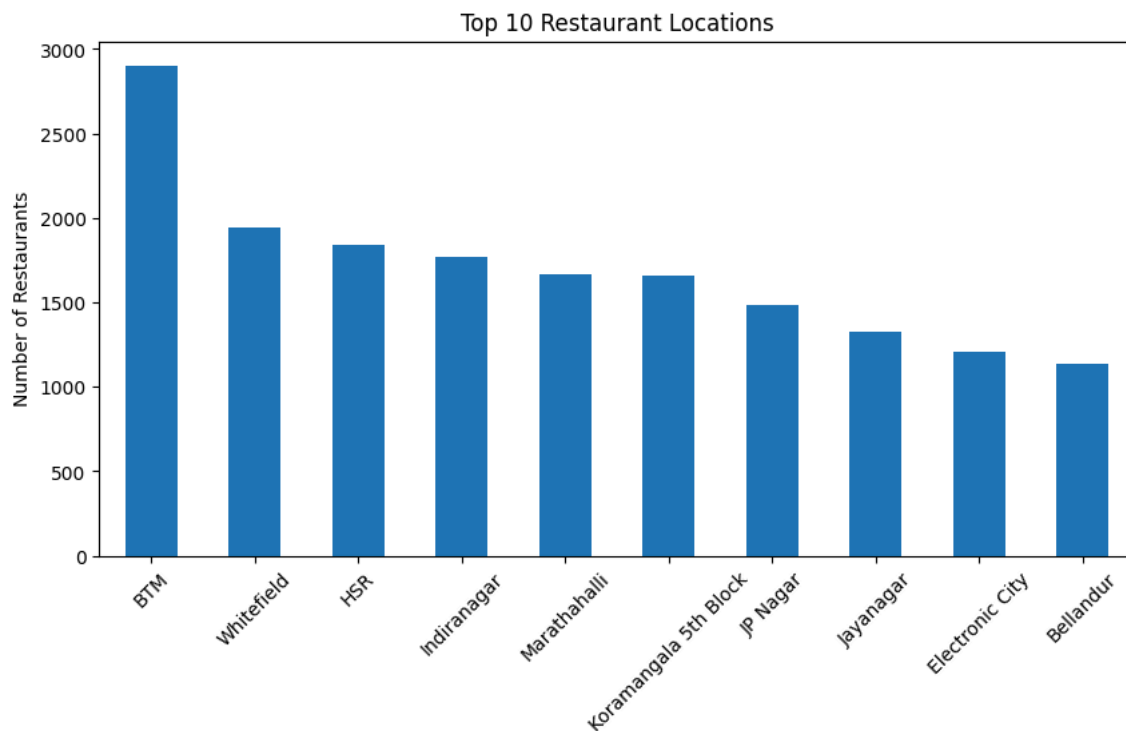
```

```
import matplotlib.pyplot as plt
```

```

top_locations.plot(kind='bar', figsize=(10,5))
plt.title("Top 10 Restaurant Locations")
plt.xlabel("Location")
plt.ylabel("Number of Restaurants")
plt.xticks(rotation=45)
plt.show()

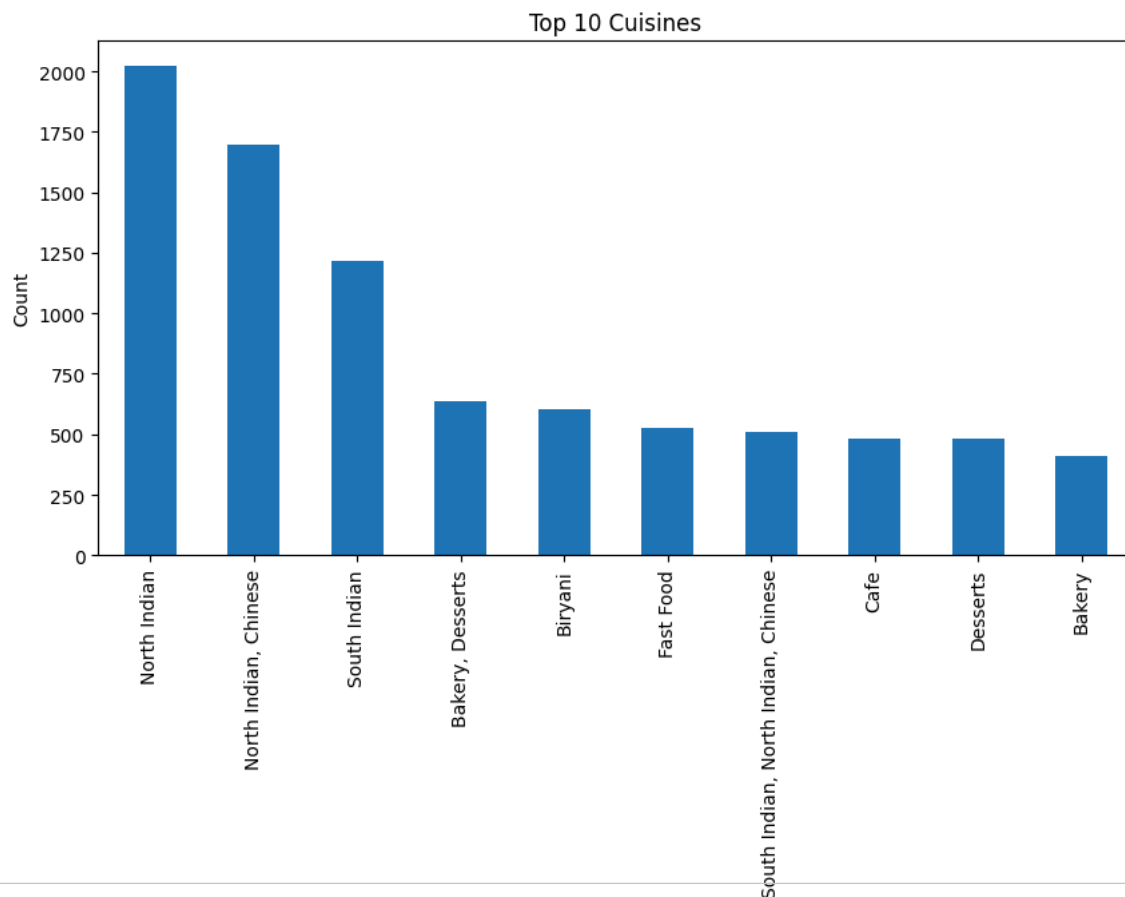
```



```
top_cuisines = df['cuisines'].value_counts().head(10)
print(top_cuisines)
```

```
cuisines
North Indian                2024
North Indian, Chinese       1694
South Indian                1216
Bakery, Desserts            634
Biryani                     602
Fast Food                   528
South Indian, North Indian, Chinese  507
Cafe                        482
Desserts                    481
Bakery                      413
Name: count, dtype: int64
```

```
top_cuisines.plot(kind='bar', figsize=(10,5))
plt.title("Top 10 Cuisines")
plt.xlabel("Cuisine")
plt.ylabel("Count")
plt.xticks(rotation=90)
plt.show()
```



```
df[['rate', 'approx_cost(for two people)']].head()
```

	rate	approx_cost(for two people)
0	4.1/5	800
1	4.1/5	800
2	3.8/5	800
3	3.7/5	300
4	3.8/5	600

```
df['rate'] = df['rate'].str.replace('/5', '', regex=False)
df['rate'] = pd.to_numeric(df['rate'], errors='coerce')
```

```
df['rate'].head()
```

	rate
0	4.1
1	4.1
2	3.8
3	3.7
4	3.8

```
dtype: float64
```

```
df['approx_cost(for two people)'] = df['approx_cost(for two people)'].astype(str)

df['approx_cost(for two people)'] = (
    df['approx_cost(for two people)']
    .str.replace(',', '')
)

df['approx_cost(for two people)'] = pd.to_numeric(
    df['approx_cost(for two people)'],
    errors='coerce'
)
```



```
df[['rate','approx_cost(for two people)']].head()
```

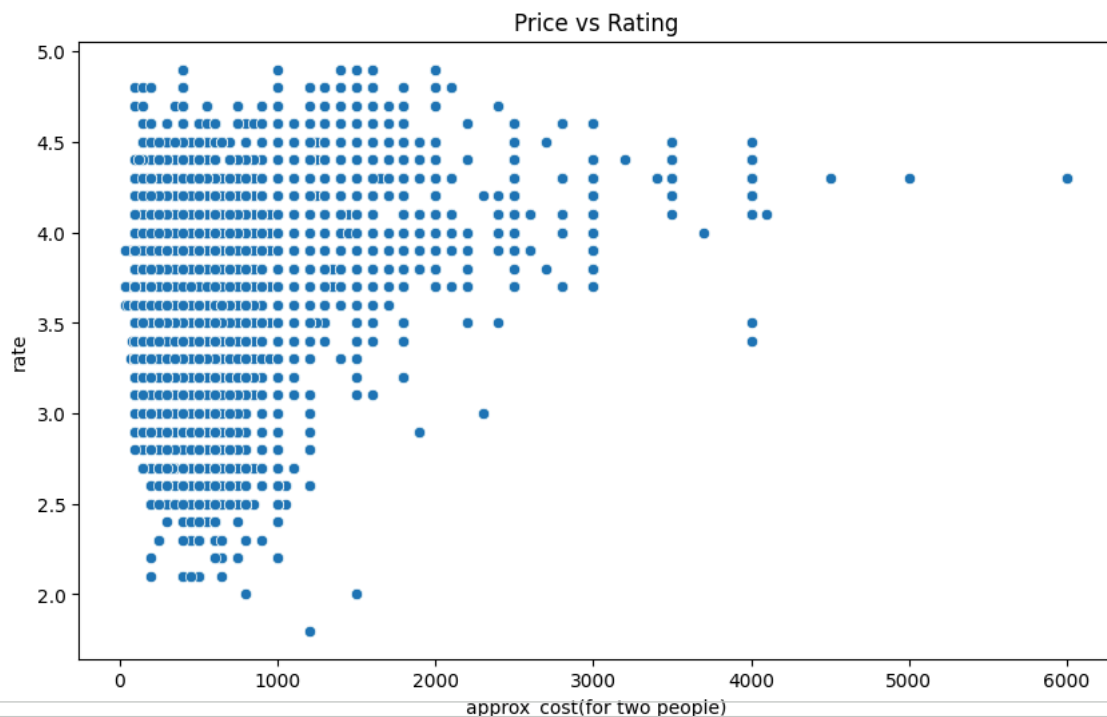
	rate	approx_cost(for two people)
0	4.1	800.0
1	4.1	800.0
2	3.8	800.0
3	3.7	300.0
4	3.8	600.0

```
import seaborn as sns
import matplotlib.pyplot as plt

plt.figure(figsize=(10,6))

sns.scatterplot(
    x='approx_cost(for two people)',
    y='rate',
    data=df
)

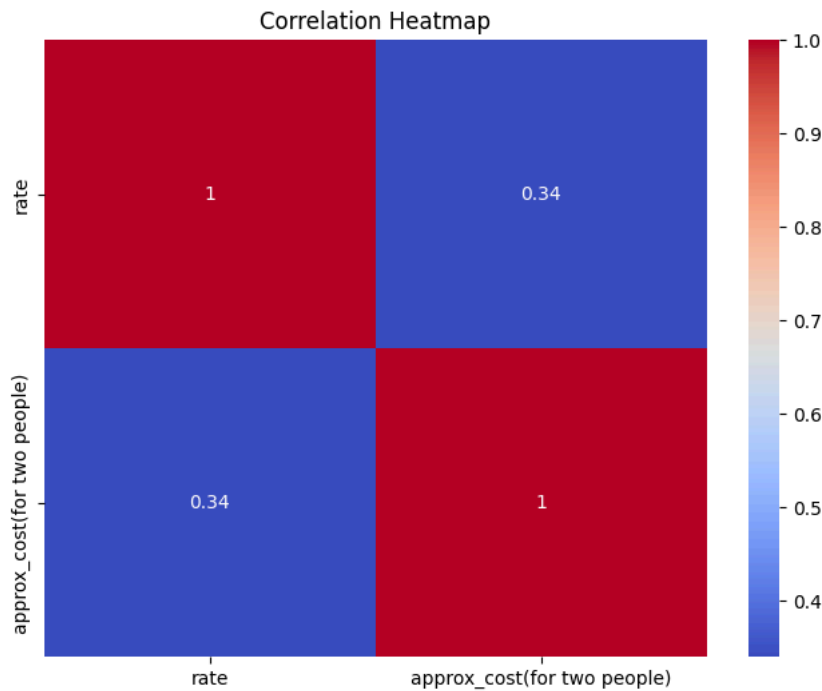
plt.title("Price vs Rating")
plt.show()
```



```
numeric_df = df.select_dtypes(include='number')

plt.figure(figsize=(8,6))
sns.heatmap(
    numeric_df.corr(),
    annot=True,
    cmap='coolwarm'
)

plt.title("Correlation Heatmap")
plt.show()
```



```
from wordcloud import WordCloud

text = " ".join(df['cuisines'].dropna())

wordcloud = WordCloud(
    width=1000,
    height=500,
    background_color='white'
).generate(text)

plt.figure(figsize=(12,6))
plt.imshow(wordcloud)
plt.axis('off')
plt.show()
```



```
top_restaurants = df.sort_values(
    by='rate',
    ascending=False
)[['name','rate','location']]

top_restaurants.head(10)
```

	name	rate	location
6117	Belgian Waffle Factory	4.9	Brigade Road
8780	Asia Kitchen By Mainland China	4.9	Koramangala 5th Block
8760	AB's - Absolute Barbecues	4.9	BTM
28270	Asia Kitchen By Mainland China	4.9	Koramangala 5th Block
55517	Flechazo	4.9	Whitefield
50220	Punjab Grill	4.9	Malleshwaram
12212	Asia Kitchen By Mainland China	4.9	Koramangala 5th Block
45826	Belgian Waffle Factory	4.9	Brigade Road

df['rest_type'].value_counts().head(10)

39947	AB's - Absolute Barbecues	4.9	BTM
	count		
	rest_type		
	Quick Bites	13050	
	Casual Dining	7988	
	Cafe	2580	
	Delivery	1674	
	Dessert Parlor	1577	
	Takeaway, Delivery	1303	
	Casual Dining, Bar	869	
	Bakery	788	
	Beverage Shop	527	
	Bar	483	

dtype: int64

```
df['rest_type'].value_counts().head(10).plot(
    kind='bar',
    figsize=(10,5)
)
```